However, to account for tied event times and survival estimates, we make the modifications listed by Ishwaran et al. (2008, Section 5.1, step 3). This is to ensure that predictions independent of $\mathbf{x}$, $\hat{S}(t \,|\, \mathbf{x}) = \hat{S}(t)$, yields $C^{\text{td}} = 0.5$ for unique event times. Note that for proportional hazards models, our metric is equivalent to the regular C-index.

### 4.2 Brier Score

The Brier score (BS) for binary classification is a metric of both discrimination and calibration of a model's estimates. In short, for $N$ binary labels $y_i \in \{0, 1\}$ with probabilities $p_i$ of $y_i = 1$, the BS is the mean squared error of the probability estimates $\hat{p}_i$, i.e., $\text{BS} = \frac{1}{N} \sum_i (y_i - \hat{p}_i)^2$. To get binary outcomes from time-to-event data, we choose a fixed time $t$ and label data according to whether or not an individual's event time is shorter or longer than $t$. Graf et al. (1999) generalize the Brier score to account for censoring by weighting the scores by the inverse censoring distribution,

$$\text{BS}(t) = \frac{1}{N} \sum_{i=1}^{N} \left[ \frac{\hat{S}(t \,|\, \mathbf{x}_i)^2 \, \mathbb{1}\{T_i \leq t, D_i = 1\}}{\hat{G}(T_i)} + \frac{(1 - \hat{S}(t \,|\, \mathbf{x}_i))^2 \, \mathbb{1}\{T_i > t\}}{\hat{G}(t)} \right]. \tag{15}$$

Here $N$ is the number of observations, $\hat{G}(t)$ is the Kaplan-Meier estimate of the censoring survival function, $\text{P}(C^* > t)$, and it is assumed that the censoring times and survival times are independent.

The BS can be extended from a single duration $t$ to an interval by computing the integrated Brier score

$$\text{IBS} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \text{BS}(s) \, ds. \tag{16}$$

In practice, we approximate this integral by numerical integration, and we let the time span be the duration of the test set. In our experiments we found that using 100 grid points were sufficient to obtain stable scores.

### 4.3 Binomial Log-Likelihood

The mean binomial log-likelihood is a commonly used metric in binary classification that measures both discrimination and calibration of the estimates of a method. Using the same inverse censoring weighting as for the Brier score, we can apply this metric to censored duration time data,

$$\text{BLL}(t) = \frac{1}{N} \sum_{i=1}^{N} \left[ \frac{\log[1 - \hat{S}(t \,|\, \mathbf{x}_i)] \, \mathbb{1}\{T_i \leq t, D_i = 1\}}{\hat{G}(T_i)} + \frac{\log[\hat{S}(t \,|\, \mathbf{x}_i)] \, \mathbb{1}\{T_i > t\}}{\hat{G}(t)} \right]. \tag{17}$$

The binomial log-likelihood can also be integrated in the same manner as (16)

$$\text{IBLL} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \text{BLL}(s) \, ds.$$

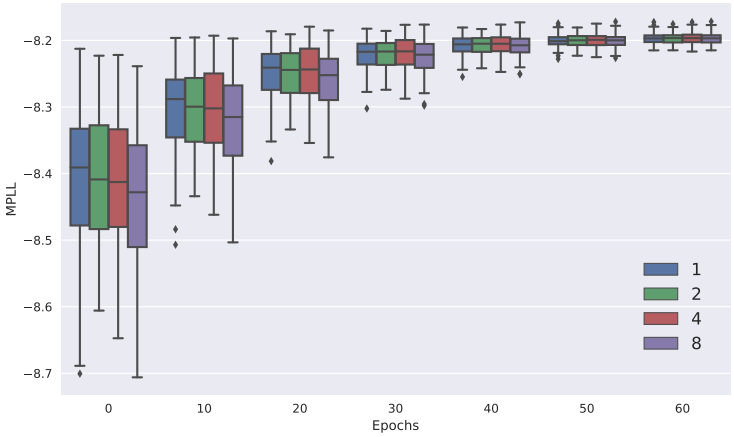



Figure 1: Box plots giving the mean partial log-likelihood (MPLL) of the test sets for different training epochs. The colors show how many controls were sampled during training (in addition to the case).

## 5. Simulations

To empirically investigate our proposed methodology, we conducted a series of simulations. These experiments are by no means exhaustive but are rather meant to verify that the methods behave as expected. In the following, we let *classical Cox* refer to a Cox regression with $g(\mathbf{x}) = \boldsymbol{\beta}^T\mathbf{x}$ obtained with the Lifelines python package (Davidson-Pilon et al., 2018). For experimental details exempt from the main article, we refer the reader to Appendix C.3.

We first investigate the behavior of our proposed loss (8). In particular, we want to examine the impact the number of sampled controls has on the fitted models, in addition to how well the results from using our loss agree with those from using the Cox partial log-likelihood, i.e., the loss (5). To this end, we simulated survival times from a proportional hazards model

$$\begin{aligned} h(t \,|\, \mathbf{x}) &= h_0(t) \exp[g(\mathbf{x})], \\ g(\mathbf{x}) &= \boldsymbol{\beta}^T\mathbf{x}, \end{aligned} \tag{18}$$

with constant baseline hazard $h_0(t) = 0.1$, and $\boldsymbol{\beta}^T = [0.44, 0.66, 0.88]$. The covariates were sampled uniformly from $[-1, 1]$. We drew censoring times independent of the covariates with constant hazard $c(t) = \frac{1}{30}$, and, in addition, we censored all individuals that were still under observation at time 30. This resulted in approximately 30 % censored individuals.

We sampled 10,000 individuals for training and 10,000 for testing, and fitted our Cox model by SGD as described in Section 3.2. This method will be referred to as *Cox-SGD*. Four models were fitted by sampling 1, 2, 4, and 8 controls (in addition to the case). The whole experiment was repeated 100 times, and the mean partial log-likelihood (MPLL) of the test sets are visualized in Figure 1. The figure indicates that the number of sampled controls does not affect the rate of convergence, but we note that the computational complexity increases with the number of sampled controls.

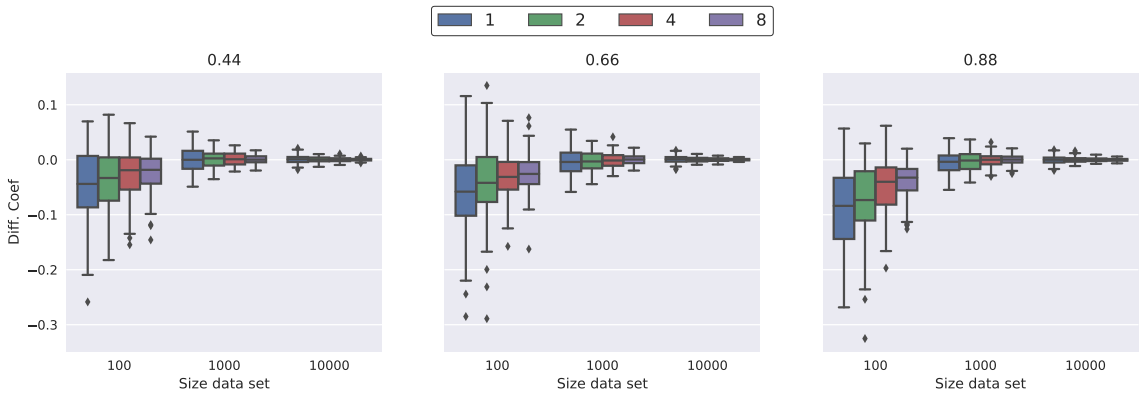



Figure 2: Differences between Cox-SGD and Classical Cox parameter estimates for different data set sizes. The numbers above the plots give the value of the true coefficient. Each box plot is based on 100 observations. The legend above the plots states the number of sampled controls for Cox-SGD.

The experiment was repeated with training sets of 1,000 individuals to verify that the results were not simply due to the size of the training data. The same patterns were found in this setting, so the figure is exempted from the paper.

Next, we compare the parameter estimates obtained from our proposed loss (Cox-SGD) with the estimates obtained with classical Cox regression. For data sets of sizes 100, 1,000, and 10,000, we fitted models to 100 sampled data sets. The differences between the Cox-SGD parameter estimates and the classical Cox estimates are displayed in Figure 2, where the legend above the plots gives the number of controls sampled for the Cox-SGD method. For the data sets of size 100, we observe that the Cox-SGD estimates seem to be slightly smaller than the Cox estimates, and this difference is larger for fewer sampled controls. However, as the data sets increase in size, the estimates for the two methods agree well.

Finally, we want to compare the likelihoods obtained by the two methods. As the mean partial log-likelihood depends on data set size, it is really not comparable across data sets. We will therefore instead use the full likelihood (2), which also depends on the baseline hazard, to compare the methods. Note that the partial likelihood may be interpreted as a profile likelihood, so the full likelihood and the partial likelihood are closely related (see, e.g., Klein and Moeschberger, 2003, p. 258). We obtain cumulative baseline hazard estimates by (6), and baseline hazard estimates by numerical differentiation of the cumulative baseline hazard estimates. We report the mean log-likelihood (MLL) of (2) to compare results for different data set sizes. In Figure 3, we show the difference in the MLL between the Cox-SGD method and the classical method for Cox regression (for each sampled data set). We used the training MLL as we are here only interested in the losses' abilities to optimize the objective, and not the generalization to a test set. From the figure, we observe that, for smaller data sets, more sampled controls in Cox-SGD seems to give likelihood estimates closer to those of a classical Cox regression. As the data sets increase in size, the MLL of the Cox-SGD seems to converge to that of the classical Cox, regardless of control sample

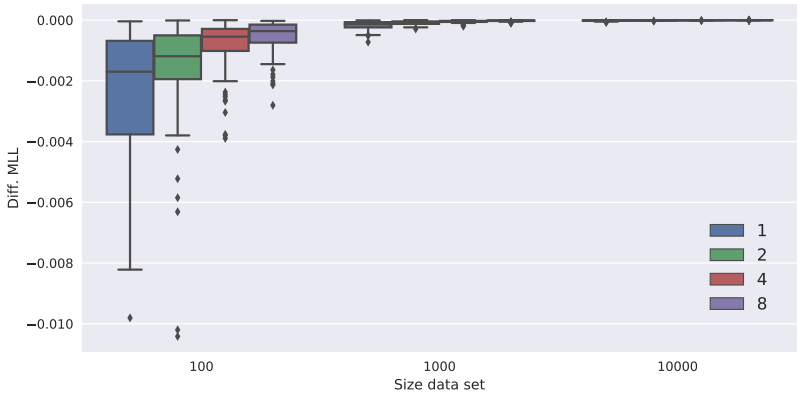



Figure 3: Differences in mean log-likelihood between Cox-SGD and classical Cox using (2). The figure gives results for different data set sizes, and the legend gives the number of sampled controls. Each box plot is based on 100 observations.

size. The MLL's of the classical Cox regression is approximately -2.2, meaning that even for the smallest data sets the differences in MLL for 1 sampled control is around 0.1 %. Hence, it seems that our loss (8) approximates the negative partial log-likelihood rather well. Furthermore, while a higher number of sampled controls can give lower training error, the effect of the number of sampled controls decreases with the size of the data sets.

In the further experiments, we will for simplicity only use one sampled control, as this was found sufficient. Furthermore, in Appendix C we have included simulations for the methods using neural networks for non-linear and non-proportional models, to verify that our proposed methods behave as expected. In summary, these simulations verify that for observations drawn from a proportional hazards model with a non-linear relative risk function, Cox models parameterized by neural networks provide better estimates of the partial log-likelihood than classical Cox models. Further, by drawing observations from a non-proportional relative risk model, the simulations verify that our Cox-Time method (Section 3.4) is able to obtain better estimates of the survival functions than methods which assume proportional hazards.

## 6. Experiments

In the following, we will evaluate our proposed methods on real data sets and compare their performance to existing methods from the literature. In total, we use five data sets. One large data set for a more in-depth analysis (see Section 6.2), and four smaller data sets commonly used in the survival analysis literature.

### 6.1 Four Common Survival Data Sets

We base this experimental section on the data sets provided by Katzman et al. (2018), as they are made available through the DeepSurv python package, and need no further prepro-